NSE Cross-Sectional Alpha Research: do classic price signals survive Indian costs and an honest holdout?

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Abstract

We ask whether eight classic daily price and volume signals predict cross-sectional returns on the National Stock Exchange of India after realistic costs, using a survivorship-free point-in-time panel rebuilt from exchange archives, a pre-registered protocol, a trial ledger and a locked out-of-sample window read once. The signals work: a LightGBM combination selected by a pre-registered rule has a walk-forward rank IC of 0.059 and a holdout rank IC of 0.041 ($t = 9.139$). The portfolios do not survive costs. A weekly long-only top-decile book beats the Nifty 500 price index gross (development IR 1.728) but, trading 29.095 times a year, loses to it net: development IR -0.917, deflated Sharpe 0.000 over 76 logged trials; holdout (2024-11-01 to 2026-10-30) IR -1.406, 95% bootstrap interval [-2.989, -0.034]. An F&O long–short book has a holdout net Sharpe of -4.448. Spreads and STT account for most of the cost drag. Every number in this paper is rendered from the run files by `qr ledger`.

1 Introduction

Do classic price signals survive Indian transaction costs and an honest holdout? Three things make this answer checkable. First, a survivorship-free daily NSE panel (3658 sessions, 3818 securities, 214 later-delisted names that were in the liquid universe) rebuilt from raw exchange files. Second, a locked protocol: a pre-registration tag before the first IC, an append-only trial ledger, and a holdout read exactly once. Third, a cost model that changes with the tax rules, including the April 2026 rise in stock-futures STT from 0.02% to 0.05%, which falls inside the holdout window.

The short answer is that the signals survive and the portfolios do not. The combined score keeps a strongly significant rank IC in the holdout, but the pre-registered weekly books turn over too fast for Indian costs, and both lose to their benchmarks after costs in the development window and in the holdout. Section 2 describes the panel, Section 3 the protocol and models, Section 4 the results, and Sections 5–6 the checks and the limitations.

2 Data

Raw files come from the NSE static archives (cash bhavcopy in the legacy and UDiFF formats, delivery files, F&O bhavcopy, index files), downloaded at no more than one request per second with an honest User-Agent and an on-disk cache. A permanent security id links symbol changes through ISIN.

Corporate actions. NSE's corporate-action export (a manual, cookie-gated download) was not used. Splits and bonuses are inferred from the data (the PRD's fallback, refined): a one-day move

above 25% whose price ratio is within 12% of a common split, bonus or consolidation ratio is treated as that action. For stocks without F&O contracts this is safe, because their price bands (at most 20%) rule out a genuine move of that size. F&O stocks have no band, so the ratio must be confirmed by the implied lot size of a futures contract alive on both days (NSE changes the lot of existing contracts only for a corporate action) or by open interest. Unconfirmed F&O moves are genuine and kept (for example SBI +28% on 2017-10-25 and Yes Bank −56% on 2020-03-06), except demerger-like days (open interest up at least 1.8× with the lot unchanged), which are nulled. Non-F&O moves that match no ratio are nulled and logged. The PRD's named checks pass (IRCTC 2021-10-28, factor 1/5; Infosys 2018-09-04, factor 1/2). The bhavcopy's own previous-close field turned out not to be split-adjusted, so it could not be used. Dividends are not captured: stock returns are *price* returns, and the long-only book is compared with the Nifty 500 *price* index.

Universe. Rebuilt each month from data dated before the month starts: EQ/BE series, at least 252 sessions of history, at least 55 trading days in 63 sessions, close at least Rs 10, median traded value at least Rs 1 crore, top 750 by value. Data-quality gates and survivorship counts are in the ledger (`data.*`).

3 Method

3.1 Signals and evaluation

Eight signals with signs and windows fixed before any IC was computed (`PREREG.md`): MOM_12_1 [Jegadeesh and Titman, 1993], REV_5, LOWVOL_63, IVOL_63 [Ang et al., 2006], AMIHUD_63 [Amihud, 2002], ABVOL_5_63, DELIV_ABN and HIGH52 [George and Hwang, 2004]. Each is winsorized at 1/99, standardized by date, neutralized to sector dummies, a shrunk Dimson beta and size, and re-standardized. We report rank IC at horizons 1, 5, 10 and 21 sessions with Newey–West t statistics [Newey and West, 1987] and Holm adjustment across signals; a signal passes at $t \geq 3$ after Holm correction.

3.2 Combination

Equal weight, ridge and LightGBM are compared on eight expanding-window folds with a 27-session purge and embargo. The headline is the better of ridge and LightGBM only if it beats equal weight with Newey–West $t \geq 2$ on the IC difference, otherwise equal weight [López de Prado, 2018, Gu et al., 2020].

3.3 Portfolios and costs

P1 is a long-only equal-weight book (enter in the top 10%, hold to the top 20%, weekly, signal at the close and fill at the next close, 10% of ADV cap) against the Nifty 500 price index (stock returns exclude dividends). P2 is long the top quintile in cash and short the bottom quintile in near-month stock futures. Costs are per side: dated STT, exchange charge, SEBI fee, stamp duty, brokerage (assumed 3 bp delivery, 1 bp futures), GST and the DP charge, plus an Abdi–Ranaldo half-spread [Abdi and Ranaldo, 2017] and square-root impact $Y\sigma\sqrt{Q/\mathrm{ADV}}$ with $Y = 1$. Table 1 lists the dated statutory and broker rates (generated from `config/costs.yaml`). Capacity is the smaller of the AUM at which the net metric halves and the AUM at which more than 20% of desired trading is blocked by the ADV cap.

Component	Segment	Side	From	Rate (bp)
stt	Delivery	both	2012-01-01	12.500
stt	Delivery	both	2013-06-01	10.000
stt	Stock futures	sell	2012-01-01	1.000
stt	Stock futures	sell	2023-04-01	1.250
stt	Stock futures	sell	2024-10-01	2.000
stt	Stock futures	sell	2026-04-01	5.000
exchange	Delivery	both	2012-01-01	0.307
exchange	Stock futures	both	2012-01-01	0.183
sebi	Delivery	both	2012-01-01	0.010
sebi	Stock futures	both	2012-01-01	0.010
stamp	Delivery	buy	2012-01-01	1.500
stamp	Stock futures	buy	2012-01-01	0.200
brokerage	Delivery	both	2012-01-01	3.000
brokerage	Stock futures	both	2012-01-01	1.000
GST on brokerage + exchange + SEBI				18%
DP charge per scrip per sell day	Delivery	sell		Rs 15.34

Table 1: Per-side cost schedule; the latest row on or before the trade date applies.

3.4 Validation protocol

A pre-registration commit and tag precede the first IC; `oos.lock` makes every loader blind after 2024-10-31; every signal, combiner fold and base portfolio is logged to the trial ledger before results print; the deflated Sharpe ratio [Bailey and López de Prado, 2014] uses the true ledger count; confidence intervals use a stationary block bootstrap [Politis and Romano, 1994].

4 Results

All values below are rendered from the run files by `qr ledger`; the ledger (`RESULTS.md`) names the file, key and git SHA behind each one. Development numbers are walk-forward (each test fold scored by a model fitted only on earlier data, with purging and an embargo).

4.1 Single signals

Eight signals were tested raw and after sector, beta and size neutralization, at horizons of 1, 5, 10 and 21 sessions. The strongest neutralized signal by Newey–West t at $h = 5$ is REV_5: mean rank IC 0.030 ($t_{NW} = 11.826$). The largest mean rank IC of any signal at $h = 5$ is 0.052, well below the pre-registered 0.08 leakage tripwire. Of the 16 pre-registered tests (8 signals, raw and neutral), 12 clear the joint hurdle of a Holm-adjusted $p < 0.05$ and $|t| \geq 3$ with the expected sign. Two liquidity-flavoured signals (Amihud illiquidity and abnormal volume) come out significant with the *wrong* sign after neutralization, so they are reported as failures of the pre-registered hypothesis, not re-signed. Low volatility, idiosyncratic volatility and short-term reversal are the strongest effects; their cumulative ICs (Figure 1) rise steadily through the sample.

4.2 Combination

The pre-registered rule picks the better of ridge and LightGBM by mean fold rank IC and keeps it only if it beats the equal-weight z-score average with a Newey–West $t \geq 2$ on the daily IC difference. The rule selected LGBM: mean walk-forward rank IC 0.059 against 0.045 for equal weight (paired difference $t = 6.887$). The headline score's IC does not decay with horizon: 0.045 at one session

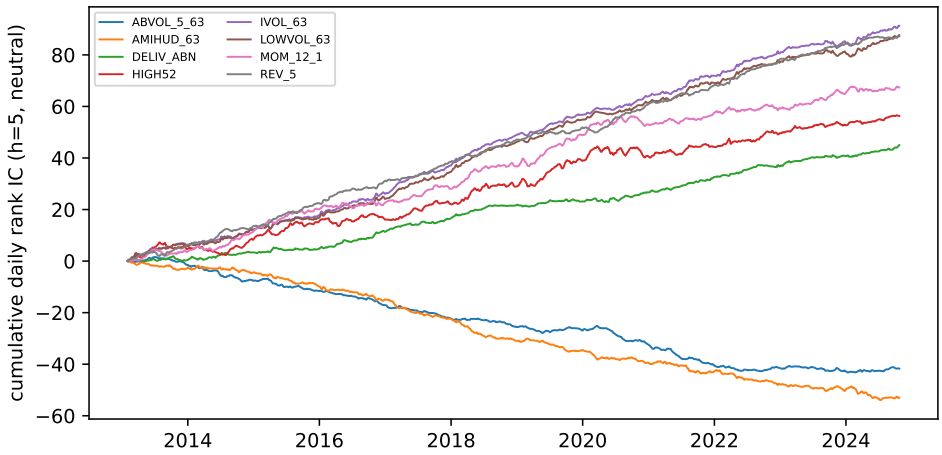


Figure 1: Cumulative daily rank IC of the neutralized signals ($h = 5$), development window.

and 0.061 at 21 sessions (Figure 2). This matters for what follows: the score's *information* is slow, but the score itself is re-ranked every week.

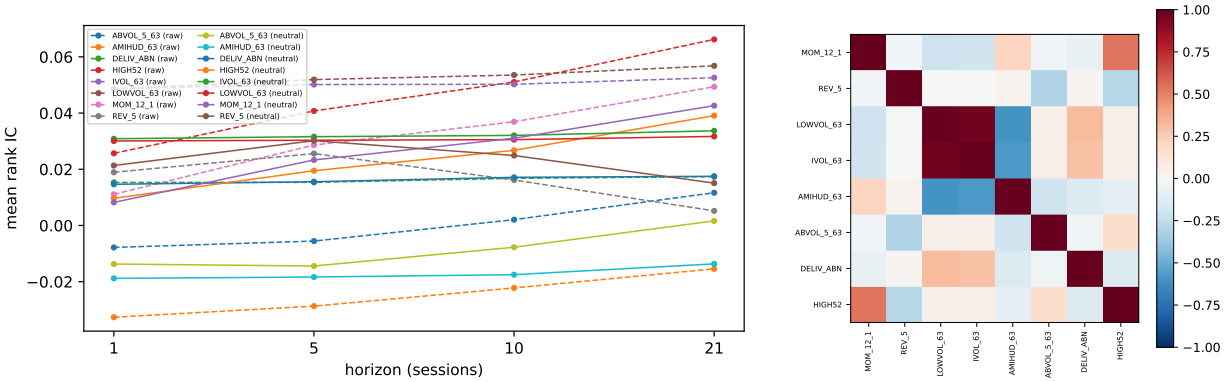


Figure 2: Left: IC decay by horizon. Right: correlation of the signals' daily scores.

4.3 Development portfolios

The gross edge is large. The decile long–short spread (P0, equal weight, no costs) has a Sharpe ratio of 2.932, and the long-only book (P1: enter the top 10%, keep while in the top 20%, at most 10% of 63-day ADV per trade, Rs 1 crore) beats the Nifty 500 price index by 0.175 a year gross (IR 1.728). Costs remove all of it. P1 turns over 29.095 times a year one-way, and the full Indian cost model drains 2690.605 bp a year: 45.400% of that is the estimated bid–ask spread, 21.627% STT and 21.036% market impact. Net of costs P1 compounds at 0.047 a year against 0.161 for the index, an active return of -0.099 (net IR -0.917). The F&O long–short book (P2) is worse: with 49.146 one-way turnover a year its net Sharpe on capital is -2.457.

	P1 gross	P1 net	P2 gross	P2 net
Information ratio (P1) / Sharpe (P2)	1.728	-0.917	0.522	-2.457
CAGR	0.370	0.047	0.118	-0.194
Max drawdown (net)		-0.676		-0.817
One-way turnover per year		29.095		49.146
Average long holdings		82.486		35.696
Cost drag, bp per year		2690.605		3320.272
PSR(0), net		0.007		0.000

Table 2: Development window 2017-01-02 to 2024-10-31 (walk-forward). P1 benchmark: Nifty 500 price index (CAGR 0.161). Risk-free: constant 6.5% fallback. P1 is evaluated against the benchmark (IR); P2 on capital (Sharpe).

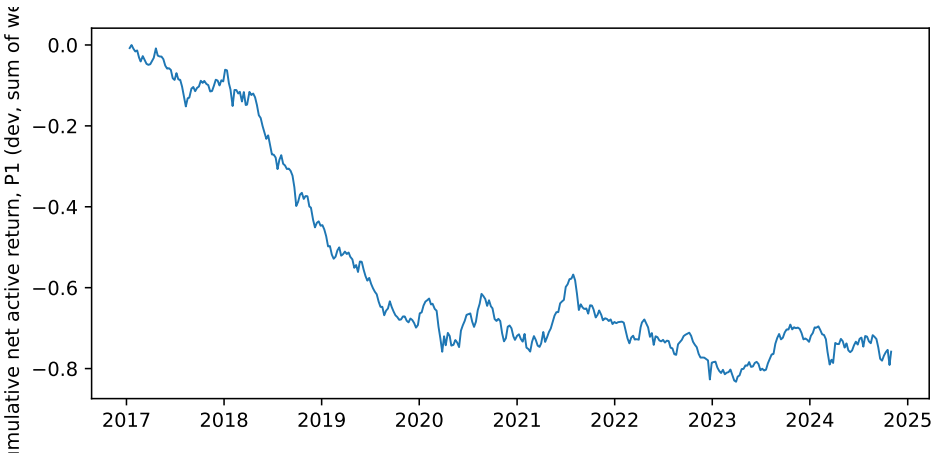


Figure 3: P1 cumulative net weekly active return against the Nifty 500 price index, development window.

4.4 The holdout read

The model and both books were frozen, rehearsed on development data, and then run once on the locked window 2024-11-01 to 2026-10-30. The signal survived: the frozen score's mean rank IC is 0.041 (t_{NW} = 9.139), and the gross decile spread has a Sharpe of 2.030. The portfolios did not. Gross, P1 beat a falling index (CAGR -0.014) by 0.080 a year (IR 0.852); net, it lost -0.141 a year against it (IR -1.406, bootstrap 95% interval [-2.989, -0.034], which excludes zero on the negative side). Costs were 2179.790 bp a year, 48.479% spread and 26.048% STT. The net beta to the index was 1.269. P2 lost money even before costs (gross Sharpe -0.246) and its net Sharpe is -4.448.

4.5 Multiple testing and capacity

Across 76 logged trials (21 effectively independent at $\rho > 0.7$), the deflated Sharpe ratio of the development P1 net returns is 0.000: once the search is accounted for, there is no evidence of a positive net Sharpe. Capacity is defined as the smaller of the AUM that halves the net metric and the AUM at which the ADV cap blocks 20% of desired trades; with a negative net metric at the smallest size it is undefined (P1 development: none (net metric ¡= 0 at Rs 1 crore); holdout: none (net metric ¡= 0 at Rs 1 crore)). Costs would have to fall to 0.615 times the modelled level for P1 to break even on the development window (P2: 0.144). The capacity curve (Figure 5) is not

	P1 gross	P1 net	P2 gross	P2 net
Information ratio (P1) / Sharpe (P2)	0.852	-1.406	-0.246	-4.448
95% bootstrap CI (net)		[-2.989, -0.034]		[-7.007, -3.663]
CAGR	0.056	-0.150	0.047	-0.229
Max drawdown (net)		-0.350		-0.422
One-way turnover per year		28.390		50.638
PSR(0), net		0.021		0.000
Cost drag, bp per year		2179.790		3033.774
STT share of drag, %		26.048		19.622

Table 3: Locked out-of-sample window 2024-11-01 to 2026-10-30, read once (block bootstrap, $B = 10{,}000$).

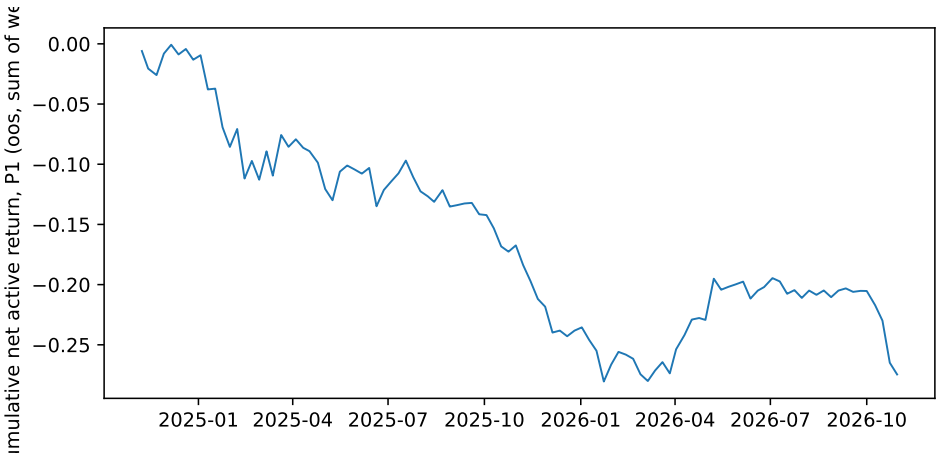


Figure 4: P1 cumulative net weekly active return in the holdout.

monotone: beyond about Rs 100 crore the net IR *improves*, because the 10%-of-ADV cap blocks an increasing share of the desired trades and so forces lower turnover. That is a second sign that the book trades more than its signal needs.

5 Robustness

The cost conclusion does not hinge on one assumption. Table 4 re-runs P1 with each cost input moved (same scores, same rules; these are sensitivities, not new strategies, and are not counted as trials). Removing brokerage entirely or halving the impact coefficient leaves the net IR negative in both windows; doubling all costs makes it much worse. A 30% haircut on names that stop trading changes little, because few held names were delisted.

Leakage checks. The real-data truncation test rebuilds the whole pipeline from raw files cut at three randomly drawn dates and requires every price, universe flag, signal and context value dated on or before the cut to be bit-identical to the full build; it passes. Neither tripwire fired (signal IC above 0.08, development net IR above 2). The first official backtest contained a real bug, a daily rather than weekly rebalance; it was found from its implausible turnover, fixed with a regression test and re-run, and its two trials remain in the ledger and in N_{all} (`DECISIONS.md` D-033).

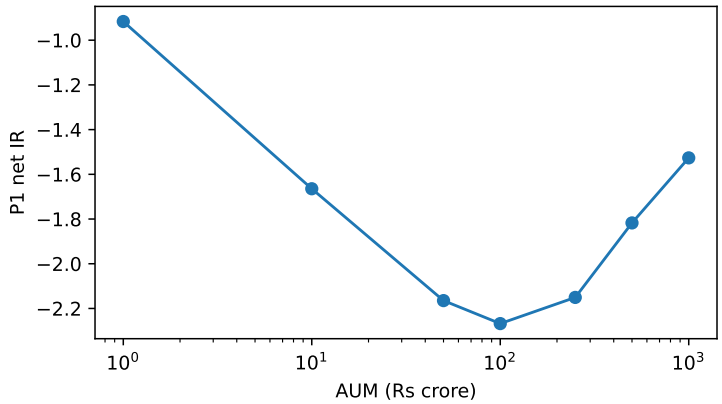


Figure 5: P1 net IR against AUM (Rs crore, log scale), development window.

P1 net information ratio	Development	Holdout
Base case	-0.917	-1.406
Brokerage 0 bp	-0.745	-1.216
Impact coefficient $Y = 0.5$	-0.662	-1.298
Impact coefficient $Y = 2$	-1.341	-1.614
30% delisting haircut	-0.932	-1.406
All costs doubled	-2.875	-3.165

Table 4: Cost sensitivities for P1 (Rs 1 crore).

Not run. The PRD's further robustness checks (no sector neutralization, a monthly rebalance, results by calendar year) were not run before the holdout was read. Running them now would be post-holdout exploration, so they are listed as future work rather than reported.

6 Limitations

- Sector labels are a 2026 snapshot, not point-in-time; names delisted earlier are `Unclassified`.
- No shares-outstanding data: size is proxied by traded value; share turnover is not tested.
- No corporate-action export: splits and bonuses are inferred by ratio matching with F&O lot/open-interest confirmation; dividends are missing (price returns; benchmark is the price index); rights issues are captured only when their ratio is near a common one; an F&O demerger without an open-interest jump would be kept as a fake move. Counts and the 50 most liquid inferred actions are in the data-quality report.
- Protocol incident (disclosed): during a plumbing rehearsal on partial development data a label shuffle silently failed, and the researcher saw the combiner comparison, one signal's IC and the turnover level before the official run. Its 33 trials are in the ledger and in N_{all}; no pre-registered choice was changed (`DECISIONS.md` D-028).
- True delisting returns are unknown; a held name carries flat for up to 10 sessions and is then written off at a 0% or 30% haircut.
- Statutory rates were spot-checked from public sources; the 2024-10-01 futures STT date and the pre-2016 delivery history are not independently confirmed (no reported number depends

on pre-2016 rates). Brokerage is assumed; futures margin is a flat 20%; lot rounding is ignored.

- The impact model is a square-root law with $Y = 1$, not fitted to Indian execution data. P2 uses cash-market ADV and volatility for the futures leg.
- The risk-free rate is a constant 6.5% fallback (no RBI T-bill series supplied).
- The trial count is a lower bound: choosing these eight signals from the literature was itself a selection the ledger cannot see. The paper's own deflated-Sharpe worked example was not reproduced (the PSR example was).
- The researcher lived through 2024–2026 markets; the lock prevents numeric peeking, not general knowledge.
- The universe is smaller than designed early on: the pre-registered Rs 1 crore median-value floor leaves as few as 276 names in a month, and 91 months have fewer than the 750-name cap, so early decile books are thin and tilted to liquid names. 2014 and 2019 have fewer sessions than the data gate expects because of election-day closures (D-030).
- Sector labels depend on the future through later renames: mapped with the security master as known at a 2014 cutoff, about a dozen universe names would be Unclassified (data-quality report; D-032). The truncation test therefore holds the mapped sector table fixed.
- The headline books trade weekly with no turnover control beyond a hold buffer. That was the pre-registered design; it is also, on this evidence, the main reason they fail.
- The holdout benchmark fell (Nifty 500 price index CAGR -0.014) and the book's beta was above one, so the holdout tests a single, weak market regime.
- A real backtest bug (daily instead of weekly rebalancing) was found and fixed before the holdout read; the commit history was later rewritten to the author's identity, which changed every SHA (map in `docs/SHA_MAP.md`).
- One market, about eight years of walk-forward plus two of holdout.

7 Conclusion

The answer to the title question is no, for the books as pre-registered. Classic price and volume signals do predict the cross-section of NSE returns, and that prediction held up in a holdout read once (rank IC 0.041, $t = 9.139$). But a weekly long-only top-decile book built on them loses to the Nifty 500 after Indian costs, in development (net IR -0.917) and in the holdout (net IR -1.406, interval [-2.989, -0.034]), and the F&O long–short book loses more. The deflated Sharpe is 0.000. This is a null result for net alpha, and it is reported as one.

The failure is specific: the score's predictive power is slow (its IC does not decay over 21 sessions), but the weekly re-ranking trades it 28.390 times a year, and spreads plus STT eat the edge. The natural next experiment is a slower book (monthly rebalance, wider hold buffer, or a turnover penalty in the optimizer), pre-registered against a new holdout, since this one has been spent.

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